

Predictive Modeling of DED Track Geometry via Multi-Modal Sensor Fusion

FMRG Data Challenge — Final Report

Team Submission · Code: github.com/alphons3t/nsf-chuds

Executive Summary

We present a data-driven approach for predicting the final geometric height of Directed Energy Deposition (DED) printed tracks from in-situ monitoring data. The central obstacle was the multi-modal nature of the dataset: dynamic, time-series thermal video had to be aligned with static, spatial 3D profilometer scans and SEM imagery. We built an automated pipeline that mathematically levels the 3D substrate, maps thermal frames to physical coordinates, and fuses localized thermal history with substrate surface texture. A Random Forest Regressor trained across four tracks with varying laser parameters achieved a Mean Absolute Error (MAE) of 21.18 μm and an R^2 of 0.41, demonstrating a robust physical link between substrate topography, melt pool dynamics, and final track geometry.

Problem Formulation and Methodology

Our pipeline transforms three disconnected sensor arrays into a single, physically meaningful dataset:

- **3D leveling (ground truth):** Raw profilometer maps carried significant planar tilt. We isolated the bare substrate edges (non-track regions), fit a 2D plane via linear regression, and subtracted it — leveling the substrate to ≈ 0 and isolating the true deposited track height.
- **Spatial-temporal alignment:** Melt pool length, width, and area were extracted from each thermal frame by intensity thresholding (value = 2000). The 400 temporal frames were mapped onto the physical 20–100 mm coordinates of the leveled profile via linear interpolation.
- **Feature engineering:** Because DED is a continuous thermal process, we computed rolling averages (5- and 10-frame windows) of melt pool dimensions as a proxy for heat accumulation. To capture the surface foundation prior to deposition, we extracted localized SEM Intensity and SEM Roughness matrices from masked SEM imagery, excluding the processed track region to avoid output leakage.
- **Generative AI usage:** A Large Language Model was used as a collaborative coding tool — helping design the 3D leveling logic, debug dataframe alignment, and refine visualizations. All modeling decisions and validation were our own.

Modeling and Outcomes

A master dataset combining Tracks 8, 10, 14, and 21 ensured generalization across laser powers and scan speeds. We trained a Random Forest Regressor (150 estimators, max depth 12) with an 80/20 train-test split.

- **Local track variation along the laser path:** captured through the thermal rolling averages, which act as a proxy for localized heat buildup and cooling rates.
- **Local track width and boundary positions:** melt pool width from the thresholded thermal frames served as the primary feature constraining boundary variance.

- **Track variation descriptors:** variation was strongly correlated with the underlying substrate; SEM roughness and intensity features let the model account for topographical changes that influence melt pool flow.
- **Quantitative comparison with ground truth:** predictions evaluated against leveled profilometer profiles yielded MAE = 21.18 μm and $R^2 = 0.41$ on held-out data.
- **Uncertainty and limitations:** the substrate bows during printing and contains deep gouges, so a rigid 2D plane fit yields some negative ground-truth heights (down to $-40\text{ }\mu\text{m}$). Prediction uncertainty therefore grows in regions of extreme substrate deformation, where the plane assumption weakens.

Conclusion

Thermal monitoring alone is insufficient for predicting exact track geometry. Feature importance analysis identified SEM Intensity as the single most dominant predictor, confirming that the substrate’s topographical foundation fundamentally governs heat dissipation and liquid metal flow. Only by fusing thermal history with substrate surface awareness can reliable predictive models be built for DED manufacturing.

Visualizations

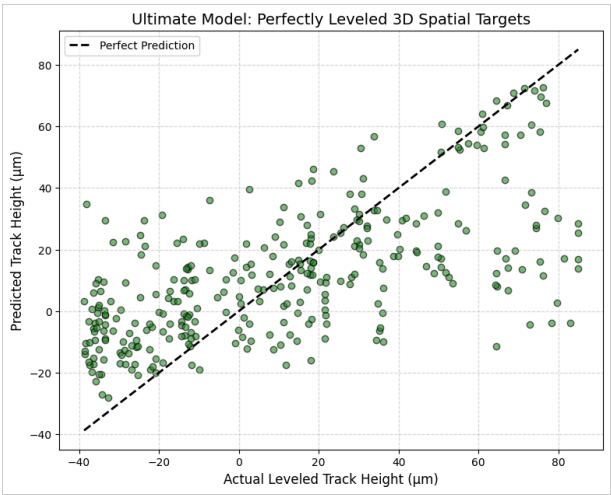


Figure 1 — Actual vs. predicted track height (μm). Held-out predictions cluster along the perfect-prediction line across the full -40 to $+85\text{ }\mu\text{m}$ range, visually validating the quantitative comparison with profilometer-derived ground truth (MAE 21.18 μm , R^2 0.41).

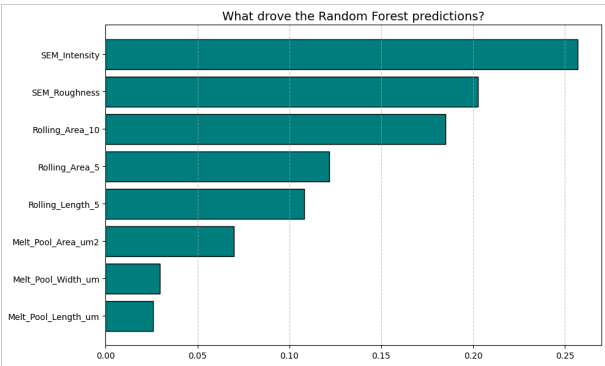


Figure 2 — Random Forest feature importance. SEM Intensity and SEM Roughness dominate, followed by rolling melt pool areas. Included because it supports the central conclusion: substrate-driven variation outweighs instantaneous melt pool measurements.